

3.0 Samples

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Overview

- Here we consider what happens when you have a sample instead of the whole population.
-

Until now we've assumed that when calculating attributes, we could do so using all of the data (i.e., the *entire* population).

However, it **may not be** possible to calculate an attribute for the population if, for example:

- the population is too large, or
- the attribute is too complex, or
- we just don't have access to the entire population 

If we have a **sample** or a subset S of $n \ll N$ units,

- * Then the attribute $a(S)$ calculated based on this sample is an estimate of its population counterpart $a(\mathcal{P})$.

$$a(S) = \hat{a}(\mathcal{P}) = a(\hat{\mathcal{P}})$$

- The second equality emphasizes that $\mathcal{S}$ is an estimate of $\mathcal{P}$.

When using a sample instead of the entire population, we might consider

- Sample error
- Consistency

Sample error

- Any difference between the actual values of the estimate $a(\mathcal{S})$ and the quantity being estimated (the estimand) $a(\mathcal{P})$ is an **error**.

$$\text{Sample Error} = a(\mathcal{S}) - a(\mathcal{P}) \leftarrow \text{difference between estimated and true values.}$$

- The nature of this error will depend on the sample and the attribute.
- Quantifying error;
 - for numerical attributes, this is determined mathematically;
 - for graphical attributes, it is not precise though still conceptually applicable.

Example - Agriculture Data

Load the data and obtain a sample of size $n = 100$

```
agpop <- read.csv("../Data/agpop_data.csv", header=TRUE)
set.seed(341)
s = sample(length(agpop$farms87), 100)
```

Handwritten notes: An arrow labeled 'N' points to 'length(agpop\$farms87)' and an arrow labeled 'n' points to '100'. The variable 's' is circled in red.

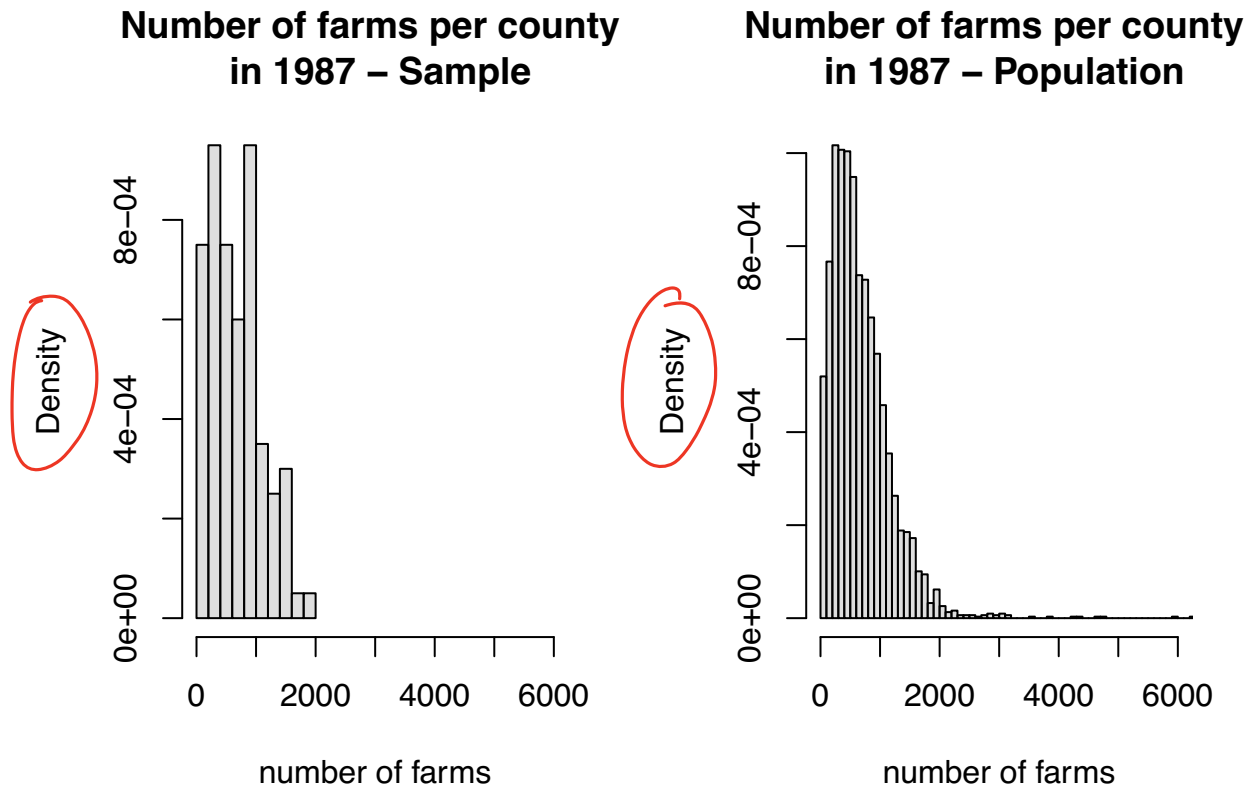
Since we have the population we can calculate the differences between some attributes.

```
c(mean(agpop$farms87[s]) - mean(agpop$farms87),
  median(agpop$farms87[s]) - median(agpop$farms87),
  sd(agpop$farms87[s]) - sd(agpop$farms87),
  IQR(agpop$farms87[s]) - IQR(agpop$farms87) )
```

```
## [1] -17.99428 74.00000 -101.08132 -12.00000
```

We can also compare the difference between histograms.

```
par(mfrow=c(1,2))
hist(agpop$farms87[s], breaks='FD', col=adjustcolor("grey", alpha = 0.5),
  main="Number of farms per county \n in 1987 - Sample",
  xlab="number of farms", prob=TRUE, xlim=c(0,6000) )
hist(agpop$farms87, breaks='FD', col=adjustcolor("grey", alpha = 0.5),
  main="Number of farms per county \n in 1987 - Population",
  xlab="number of farms", prob=TRUE, xlim=c(0,6000))
```



- For obvious reasons, an attribute with lower sampling error is preferable.

Fisher Consistency

- If the sample $\mathcal{S}$ is equal to the population $\mathcal{P}$ then the sample error should be zero (or non-existent), i.e.
$$As \ n \rightarrow N \quad a(\mathcal{S}) \rightarrow a(\mathcal{P}) \quad \text{i.e., sample error} \rightarrow 0$$
- This would mean that the estimation is in some sense **consistent**.
 - This type of consistency is sometimes called **Fisher consistency** in the statistical literature,
 - Named after the statistical scientist Ronald A. Fisher who in 1922 identified this consistency as an important criterion for estimation.

Example - Agriculture Data

Consider what happens to the sample errors of the mean, median, standard deviation and interquartile range as $n \rightarrow N$.

$n = 1000$:

```
set.seed(341)
s <- sample(length(agpop$farms87), 1000)
c(mean(agpop$farms87[s]) - mean(agpop$farms87),
  median(agpop$farms87[s]) - median(agpop$farms87),
  sd(agpop$farms87[s]) - sd(agpop$farms87),
  IQR(agpop$farms87[s]) - IQR(agpop$farms87) )
```

```
## [1] 25.73772 13.50000 38.97308 1.75000
```

$n = 2000$:

```
set.seed(341)
s <- sample(length(agpop$farms87), 2000)
c(mean(agpop$farms87[s]) - mean(agpop$farms87),
  median(agpop$farms87[s]) - median(agpop$farms87),
  sd(agpop$farms87[s]) - sd(agpop$farms87),
  IQR(agpop$farms87[s]) - IQR(agpop$farms87) )
```

```
## [1] 13.338724 18.000000 6.984866 -9.250000
```

$n = 3000$:

```
set.seed(341)
s <- sample(length(agpop$farms87), 3000)
c(mean(agpop$farms87[s]) - mean(agpop$farms87),
  median(agpop$farms87[s]) - median(agpop$farms87),
  sd(agpop$farms87[s]) - sd(agpop$farms87),
  IQR(agpop$farms87[s]) - IQR(agpop$farms87) )
```

```
## [1] 1.725724 1.000000 2.811939 -1.000000
```

```
n = N = 3078:
```

```
set.seed(341)
s <- sample(length(agpop$farms87), 3078)
c(mean(agpop$farms87[s]) - mean(agpop$farms87),
  median(agpop$farms87[s]) - median(agpop$farms87),
  sd(agpop$farms87[s]) - sd(agpop$farms87),
  IQR(agpop$farms87[s]) - IQR(agpop$farms87) )
```

```
## [1] 0.000000e+00 0.000000e+00 5.684342e-13 0.000000e+00
```

The Sample as a Population

- In every respect the sample could be considered a population itself and might even sensibly be called a “sample population”.
- Even so, we avoid this nomenclature because it flies in the face of traditional statistical language and common English usage.
 - * However, in some applications (e.g. the bootstrap, which we will discuss later), we use the term “pseudo population” in reference to the sample.
- Nevertheless, treating $\mathcal{S}$ as a population allows us to evaluate any population attribute on the sample in the same way we would for $\mathcal{P}$.
- Some samples will have a small sample error and some will have a large one.
 - To quantify this we could look at all possible samples of size n .

Summary

- Sample error is the difference between a sample estimate and the attribute calculated on the entire population.
- An attribute is Fisher consistent when the sample error is zero when the sample is the population.